Milestone 5- Team Four

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1 Abstract

The idea of toppling dominoes has been looked at many times from many different viewpoints over the past several decades. We started by developing a theoretical solution that is able to predict the fall times of certain arrangements of dominoes. In addition, multiple experiments were run so that the theoretical values can be compared and validated. Approximately every month there was a competition with other groups that are working on the same project to determine whose theoretical model was better and then we were ranked accordingly.

2 Introduction

The task of looking at a set of dominoes cascading is often seen as an elementary operation. However, when examined more closely many important questions arise. How fast are these dominoes falling? What was the initial action that produced this response? When, if at all, will the dominoes stop falling? In our case we will be looking at the central question of how to place the dominoes so that once triggered they fall at the fastest rate. Along with this question we want to note how the "topple time" varies as a function of spacing and how the arrangement of the dominoes effects this time as well. Using the equipment found in the MEC lab along with the literature provided experiments were conducted and analyzed in order to help answer these tough questions.

After looking at several different journals and other publications it became clear that there was already a vast amount of information concerning the cascading of dominoes. Some of the theory that has been developed in this area is reasonable while other theory seems to be way off base. As with any experiment there are assumptions and definitions that must be established to obtain results which one thinks directly relate to the subject matter.

Molecule cascades, although different, relate to the domino effect very closely. In a research article written by A.J. Heinrich entitled Molecule Cascades[3] there is a discussion of how there is a toppling affect that takes place on things at the molecular level as well. An experiment was conducted in which CO molecules were set up in a staggered formation and then "toppled" by triggering them with a scanning tunneling microscope (STM). Similar to dominoes one molecule after another transferred energy to its adjacent molecule thus causing it to hop to the next molecule. How much energy it takes to cause each molecule to hop is not known but it is believed to vary.

This brings about the question of what causes the amount of energy needed to vary from one domino to another. One thing that was noted in their research was that as the temperature was changed the energy of each molecule changes and thus has a drastic effect on how each molecule hops related to another. This would be expected when you consider that the parameters for which the experiment is being conducted are

changing. Although there are similarities to the domino cascade it is important not to take the results done at a molecular level and assume that they will necessarily coincide with the ones being conducted,[3].

Another example of a domino effect, is the spatial pattern of dead trees in forest. It was found that dead trees had a much higher frequency in areas close to open gaps caused by previously dead trees. This is because the conditions that caused the first trees to die are now affecting the trees near the gap that is created. These conditions are wind, soil and age. The trees along the edge of the dead tree gap are more prone to falling because of exposure to the previous conditions. This can be compared to trees along the edge of the forest itself. These trees have a more normal frequency of falling even though they are more exposed than other trees in the forest,[4].

A more closely related study was documented by Charles W. Bert entitled Falling Dominoes. Bert wished to study the central question of "How fast do dominoes fall?". Although unrelated in some sense, Bert discusses an example of how Southeast Asia follows the domino effect in that as one city fell the next soon followed. In theory it would seem that as city after city fell they would begin to fall at a faster rate. This brings about an interesting question concerning the dominoes. Do they fall at equal rates or do these rates increase/decrease exponentially? In order to answer this question many assumptions were needed. These assumptions are listed as follows:

- All of the dominoes are identical.
- The dominoes are parallel to each other and equally spaced
- There is no sliding of the dominoes on the plane and each domino remains in contact with the plane at least along one edge.
- Each domino rotates in one direction only.
- There are no external friction loses.
- The dominoes are perfectly rigid with no energy loss at contact
- The time of propagation for each domino is the same, i.e., there is no interaction between dominoes other than initial contact.

It was important to use some of these assumptions but not all of them in order to obtain the best model for predicting the speed at which the dominoes are falling. Any assumption that were calculated rather then assumed were not neglected. Charles Bert notes that some of the inconsistency of this experiment was due to the inaccuracy of their hypothesis. The largest sources of error were associated with the friction between the domino and plate. These friction forces must be accounted for in our model in order to reduce this error that Bert calculated,[2].

Similar to Bert, D. E. Shaw was interested in conducting an experiment to measure the time it took for a set up of dominoes to fall that was equally spaced. This experiment is closely related to the type of experiment that we conducted in the MEC lab over the next several weeks,[9].

Two timing gates were set up to measure the starting time as well as the ending time for the whole chain of dominoes to fall. Also, to ensure that the dominoes don't slide this experiment was conducted on a rough surface (ours will be sandpaper). This somewhat eliminates the error that Bert found to be such a problem in his experiment.

Just as in most experiments conducted with falling dominoes, energy conservation was used to model the various times at different intervals. This first results in both angular momentum of the domino as well as angular velocity. The angular velocity values at different positions is then integrated resulting in a value for the total time.

Shaw's experiment confirmed that dominoes set up in a straight line equally spaced produce linear results for the total topple time. However, it is important to note that this was only true for a set of dominoes greater then six. In order to fully understand what is going on, one must ask themselves why this is the case. Shaw developed a theory that when there is fewer then 6 dominoes the number of dominoes that are involved in the changing the energy of the system is varying but when you use more then 6 dominoes the number is constant,[9].

As with any experiment there were still sources of error that were noted by Shaw. The main area in which the experiment needs to be improved is the fact that it does not account for any friction between dominoes at the point of contact. Another thing to note is whether or not the dominoes are coming in contact with one another more then once. If so the spacing between dominoes may need to be altered and the test rerun.

A lot of new information was gained when reviewing two of W. J. Stronge's articles[10, 11]. The idea of energy balance was clearly represented with equations as well as text. In his article entitled The Domino Effect: A Wave of Destabilizing Collisions in a Periodic Array, Stronge notes that the kinetic energy required to knock over the first domino is all the outside energy needed for the entire array to fall. Momentum is transferred from one block to the next through the collisions that are taking place. Stronge reinforces this idea in The Domino Effect: Successive Destabilization by Cooperative Neighbors when he states "This cooperative group applies pressure on the block at the wavefront; a pressure that depends on the number of members that are actively involved in toppling this leading block." Once the domino rotates past an angle

$$\theta = \arctan(\frac{h}{L}) \tag{1}$$

it is no longer stable and the system is destabilized (where h and L are the height and the length of the domino). As the first domino is toppled the kinetic energy of the system increases and the ratio of energies can be related as follows:

$$\frac{\Delta P_i}{K_{0i}} = \frac{2\omega^2}{\dot{\theta}_i^{\,2}}[\cos\psi - \cos\theta] \tag{2}$$

where ω is the natural frequency of the block and K_{0i} is the initial kinetic energy of the domino. The natural frequency ω, can be represented as

$$\omega = \sqrt{\frac{3g\cos\theta}{2L}} \tag{3}$$

This change in potential energy is what controls the speed of the toppling system. After the domino becomes unstable it continues to rotate through an angle ψ until it comes in contact with the next domino. The primary speed that the dominoes topple at is controlled by a number of things including friction.

Stronge argues that when friction is accounted for, the steady state speed of the system is maximum when the dominoes are spaced far apart. After this maximum point the speed will begin to decrease again. Each collision that takes place causes a loss in kinetic energy. However, this energy is transferred to the following dominoes thus allowing the assortment to topple.

Some of the losses in energy that occur are tabulated through the coefficient of restitution. This coefficient along with the spacing of the dominoes and sliding friction are what controls the topple time. In order to have the system propagating at a constant rate it is important to use a larger number of dominoes.

Occasionally, when the friction and coefficient of restitution is large there can be a reversal in the rotation of the domino. This only happens later in the propagation and decreases the rate at which the dominoes topple. Another item to note is that often the dominoes will collide with one another more then once. This bouncing effect causes the calculated values for speed of propagation to increase and typically occurs when the dominoes are spaced very close together.

2.1 Assumptions

Many assumptions are made when coming up with a good model for cascading dominos. These assumptions are made both in the theory developed as well as the experiments conducted.

In order to simplify equations when developing a theoretical model it is important to neglect many outside forces acting on the domino. By assuming that these forces have little effect on the overall system many terms can be dropped. Also, many aspects concerning the dominos themselves are assumed to be constant from one domino to the next. Several of our key assumptions are as follows:

- All of the dominos are identical (dimensions, weight, surface roughness)
- The dominos will be equally spaced and parallel to one another when tested
- There is no sliding of the dominos on the plane and each domino remains in contact with the plane along at least one edge
- Each domino rotates in only one direction
- There are no external friction loses
- The dominos are perfectly rigid with no energy loss at contact
- There is no interaction between dominos other then the initial contact

3 Experimental

In order to check and see if the theory developed is accurate, it is important to conduct several tests. The experimental values for topple time will vary from the theory for several reasons. Often times it is important to simplify the equations being used so that they can be solved. Eliminating several variables allows us to solve for the topple time without conducting a single experiment but it is not always accurate.

Neglecting forces such as friction and drag may drastically simplify our theory, but results in erroneous values. By comparing our theory with these experiments were able to judge how large this error is, and what may be causing it. However, it must be noted that experimental results contain error as well.

The first thing that needed to be done before testing could begin was to establish a method for which all tests would be conducted. The various spacing of the dominoes was kept constant by taping a ruler down to the table and lining the center of each domino up with the appropriate "tick" mark. Because slippage was a concern, a piece of sandpaper was tapped to the table. A sample of thirty dominoes was taken and it was observed that the masses varied drastically. Thus, an average (4.7 grams) was taken and only dominoes which fell within a range of plus or minus one tenth were kept. This was done in order to keep our system as consistent as possible. It was noted that not all of the dominoes were completely vertical when stood up and thus when possible, these dominoes were discarded. To avoid error the dominos were arranged in the same order for each trial, simply adding additional dominoes as the system grew.

The next step was to set up a measuring tool known as a photogate, to record the time of initiation as well as the final topple time. Two Pasco Scientific ME-9498A Photogate Heads were chosen. Both of these measuring tools consist of a sensor which is motion sensitive and an adapter which must be plugged into the appropriate port to record data. As the domino begins to topple the sensor is triggered and the timer is started. Similarly, the second photogate is triggered as the domino crosses the sensor and the timer can be stopped. In order to achieve the closest values possible, the first gate was set up at a position located right at the top of the domino just before toppling occurs.

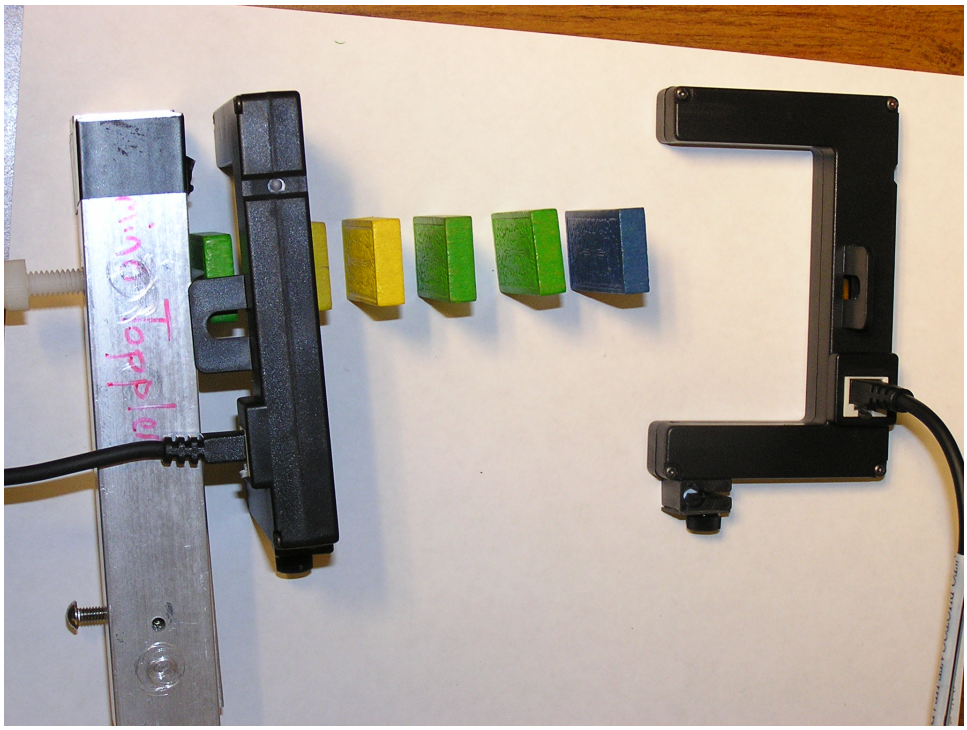


Photo 1: Photogate setup

It was also important that the initial domino not have any angular velocity before reaching this topple position. So a mechanism was created to initiate this process. Using a screw, the domino was pushed forward until it reached its balance point. The first photogate was elevated using a ring stand and lined up at this position. Next, the second photogate was set at the end of the string of dominoes. The distance for which the final domino would need to fall in order to trigger the photogate was determined by lining up another domino at the end of the grouping. Similar to the first photogate the position was set by taping the device to the table. However, unlike in the initial photogate this sensor would be set flat on the table so that the time would be recorded just as the domino struck the table.

The photogates record the time when the specimen enters the beam as well as when it leaves. After thinking about this for a while, we decided that it would be most accurate to take the time entering the beam for photogate one (0V) and the time the specimen leaves the beam for photogate two (5V). The reason that it is important to take the time the domino leaves the beam for photogate two rather then when it enters is simply because the domino will be closer to the table at this point thus completing the topple. Finally, the computer program, Data Studio, was opened and we were ready to being testing.

For each desired spacing tests were run with a string of six, seven, eight, nine, and ten dominoes. Due to the relevant angle at which the dominoes begin to topple it was necessary that the shortest span be set at 1.5 centimeters. Spans of 2.0 and 2.5 centimeters were also investigated. Each of these tests was run at least three times in order to calculate a relevant average. If any particular trial yielded a time which was much different from the others more tests were run and this value was discarded. The following averages were noted:

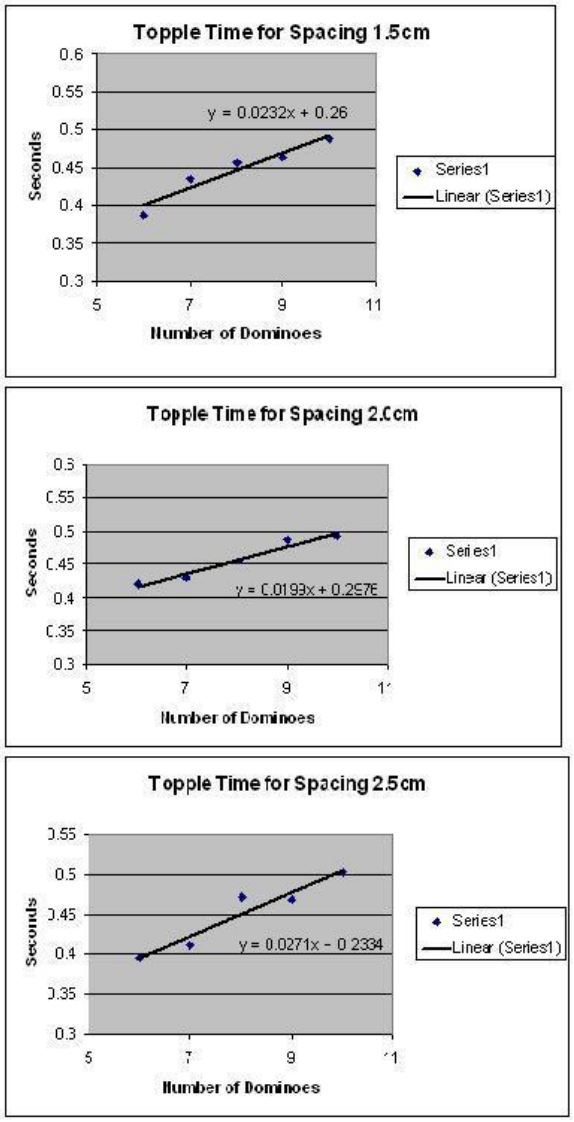


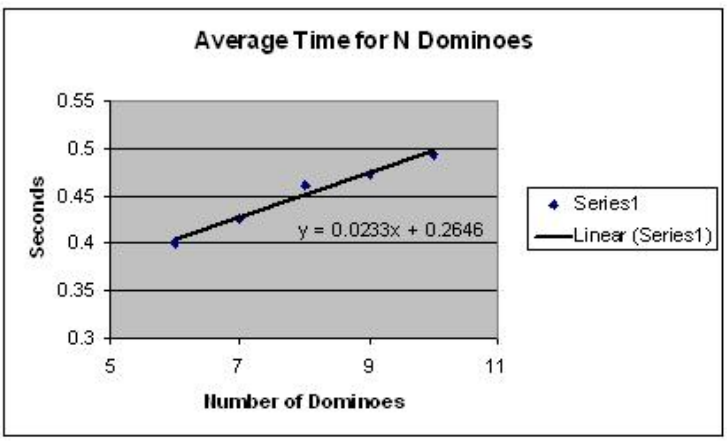
Chart 1: Experimental Results for Topple Times at various equal spacing with linear fit

As expected the topple times increased as the number of dominoes grew larger. It seemed as though the times varied linearly as the number of dominoes increases when looking at a single spacing. This can be seen through a linear fit of the Time vs. Dominoes graphs.

Topple Times					
	6 dominoes	7 dominoes	8 dominoes	9 dominoes	10 dominoes
1.5 cm	0.386	0.435	0.456	0.463	0.488
2.0 cm	0.420	0.431	0.455	0.486	0.492
2.5 cm	0.396	0.412	0.471	0.469	0.503

Chart 2

A closer look at the data reveals an even more interesting piece of information. As the span increases there is no large variant of time from the previous spacing. Although this may be due to experimental errors this is worth noting. When graphed simultaneously it is obvious that these curves do exhibit very similar behavior regardless of spacing.



Graph 1

This revelation led us to the idea of developing a single linear equation to predict the topple time of any spacing within the appropriate length of the dominoes. In order to do this we needed to calculate an average topple time for all domino combinations between six and ten. These averages were then graphed and a linear fit was performed.

With "X" and "Y" being the number of dominos and topple time respectively, the following linear equation can be used to estimate topple times based solely on our experimental values:

$$Y = 0.0233X + 0.2646 \tag{4}$$

Although this equation will not give us a precise topple time for any combination of dominoes and spacing it should put us close when in the range of 1.5 to 2.5 centimeters. Any span less then 1.5 centimeters will apply a pressure on the following domino and thus should be discarded. Similarly, any span which is chosen to be much larger then 2.5 centimeters will not work because the initial domino is not long enough to trigger the cascade.

As with any experiment there are always sources of error that must not be neglected. Most notably is simple human error. This encompasses a large range of items. One such item would be any measurement errors that may occur during testing. This includes the span not being uniform and the final photogate being located at the edge of where the domino will fall. There is also error associated with the positioning of

the first photogate at the exact topple position. If the sensor is too close the domino will trigger it before it falls thus extending the overall topple time. The reverse would be experienced if the sensor was too far back. Lastly, although small, there is error associated with the photogates themselves. When recording data, the sensors can only be as accurate as they have been designed.

Although useful, the previous results only go so far in analyzing the cascade effect on a string of dominos. All prior tests and experiments were conducted at a given spacing and with a set number of dominos. It was also desired that a model be developed which will minimize the topple time over a given length with an infinite number of dominos. Earlier experiments only provided information for cascades where the spacing was held constant. New theories and experiments have been constructed to answer the central question of whether or not a constant spacing provides the shortest topple time.

After noticing a considerable amount of error when conducting experiments with the photogates, a new measuring technique was employed. A *Fastcam-512PCI* high speed camera was set up to view the toppling effect in slow motion. The high speed camera was chosen for several different reasons. First, there would be no intricate setup of the photogates thus reducing human error. The trial and error process of lining the photogate up precisely was time consuming and not always correct. Also, if the photogate was moved the slightest bit, all of the tests must be redone to ensure that all of the results were collected under the same conditions. Using the high speed camera allowed us to slow down the cascade of dominos and remain consistent with when the start and stop time were collected. Using *Photron Motion Tools* we were able to view the entire toppling effect and calculate accurate start and stop times for the falling dominos.

Original tests were conducted with the same screw and toppling device that was used with the photogates. Zooming in on the screw as the domino began to fall was giving inaccurate results at first. It seemed as though the human error involved with deciding when the domino left the screw was so large that it would be impossible to get accurate results with the current set up. In order to correct this error the plastic screw was sharpened to a point using a razor blade. This was done so that it would be much clearer as to when the screw and domino were no longer in contact and thus the start time could be documented. It was noted that you could zoom in far enough and see the exact contact point between the domino and screw through only one pixel on the screen. Conducting the experiments this way allowed us to collect very repeatable data. Another technique used to eliminate this human error was the removal of dark colored dominos. The blue and green dominos were not used because they did not show up nearly as well on the video screen thus making the start time very subjective. The end of the cascade was not nearly as subjective to human error and could easily be repeated from experiment to experiment. Using the *Photron Motion Tools* a line was constructed which extended from the corner of the last domino directly out onto the surface where the toppling would be complete. As soon as the domino hit this line the time was noted and the toppling was complete. However, it was interesting to watch how the dominos bounced back up and in some cases caused a momentarily reversal of cascade. These videos turned out to be useful in making assumptions that helped simplified our theoretical model. One area in which the photogates proved to be better then the high speed camera was the range of data that could be taken. The high speed camera allows for a viewing field of only about twenty five centimeters, where as the photogates can collect data of any range. Due to this limited viewing range all tests were conducted between five and twenty five centimeters.

The precision involved in calculating the topple time is a time consuming process and thus not as many tests were conducted for each scenario. Often this would result in erroneous data but a few introductory tests proved the camera to be very consistent from trial to trial. Three experiments were run to calculate the topple time of a single domino and the results are as follows:

Single Domino Topple Time

Start Time(s)	Stop Time(s)	Topple Time(s)
0.666	0.967	0.301
0.618	0.914	0.296
0.655	0.955	0.3
	Average=	**0.299**

Chart 3

It can clearly be seen that data varies only slightly from trial to trial. The three topple times noted differed by only five thousandths of a second which led to the conclusion that one trial was substantial for all of the subsequent experiments.

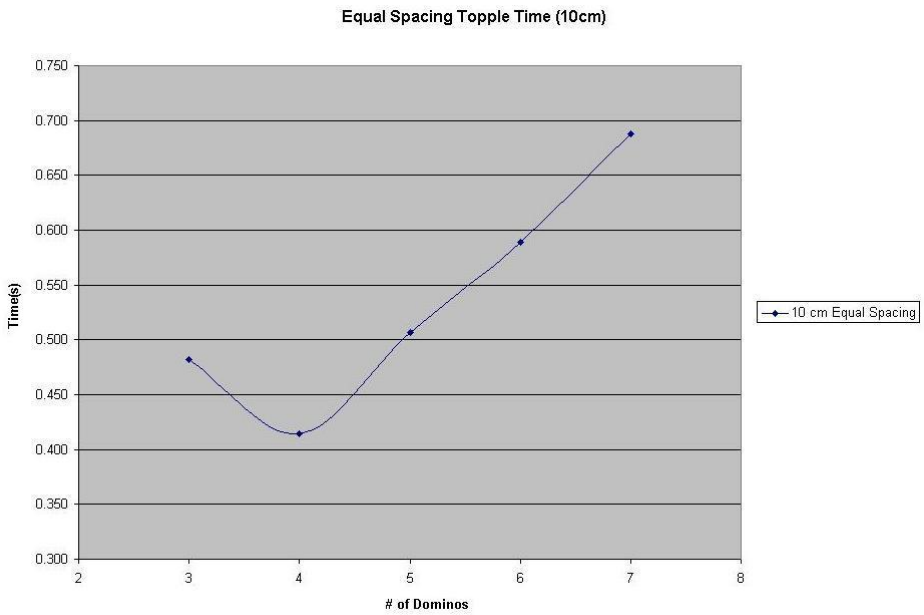
Before beginning any experiments we felt it was necessary to make a prediction as to which arrangement would provide the fastest topple time. Initial guesses were that with an equal spacing the dominos would cascade at a faster rate then with any other assortment of dominos. If this hypothesis proves to be true we then must optimize the system for several different lengths and see if a pattern develops.

For lengths of five, ten, fifteen, twenty, and twenty five centimeters any possible assortment of dominos was set up with equal spacing between one another. All spacings were set up in the same manner. The front face of the first domino was placed at the starting position and the back face of the last domino was set at the end position. The equation used to calculate each of these spacings was the following:

$$S = \frac{L - 0.7}{N - 1} \tag{5}$$

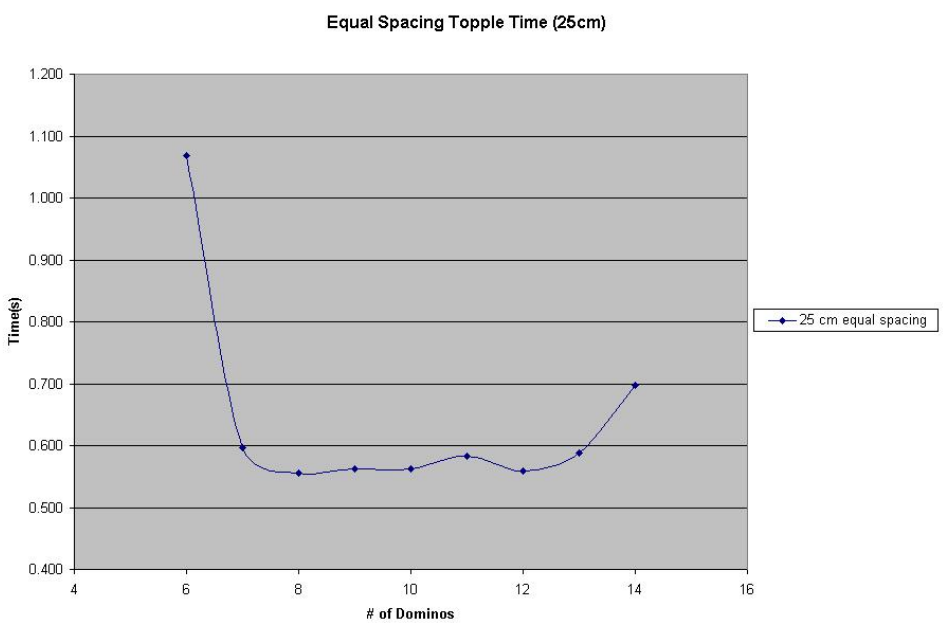
- S = Spacing
- L = Total Length
- N = Number of Dominos

As expected different spacings proved to minimize topple time depending on the overall length of our system. It seemed as though for shorter lengths, such as five centimeters, larger spacings were beneficial, where as with longer lengths smaller spacings were ideal. This ideal spacing can be clearly seen through the minimum point on the Dominos vs. Topple time graph for ten centimeters.



Graph 2

Similar graphs were noticed for all other spacings between five and twenty centimeters. However, the data for twenty five centimeters seemed to argue with the fact that as you increase the length you also increase the number of dominos used to minimize topple time. Several of the data points on this graph were very similar and fluctuated up and down rather then coming to a specific minimum value.



Graph 3

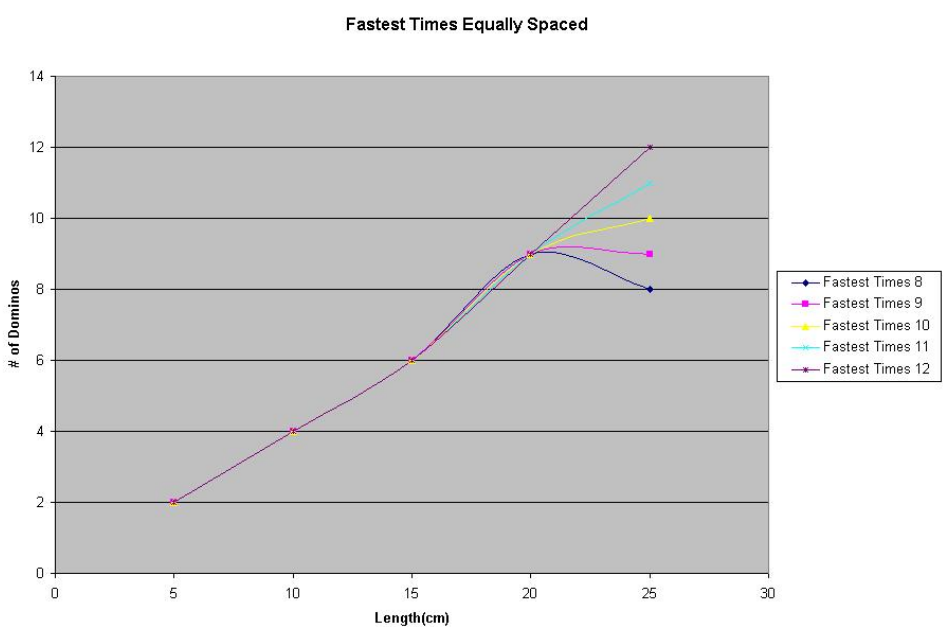
Several factors may account for why this graph differs so greatly from the others. Because all the data

points are so close it is tough to say exactly what this minimum point would be if not for human error and maybe the graph isn't as far off as originally thought.

Length(cm)	# of Equally Spaced Dominoes	Spacing(cm)	start time	stop time	fall time
25	6	4.86	2.300	1.232	1.068
25	7	4.05	1.745	1.148	0.597
25	8	3.47	1.965	1.410	0.555
25	9	3.04	1.660	1.098	0.562
25	10	2.70	1.695	1.133	0.562
25	11	2.43	2.300	1.717	0.583
25	12	2.21	2.083	1.524	0.559
25	13	2.03	1.660	1.072	0.588
25	14	1.87	2.149	1.451	0.698

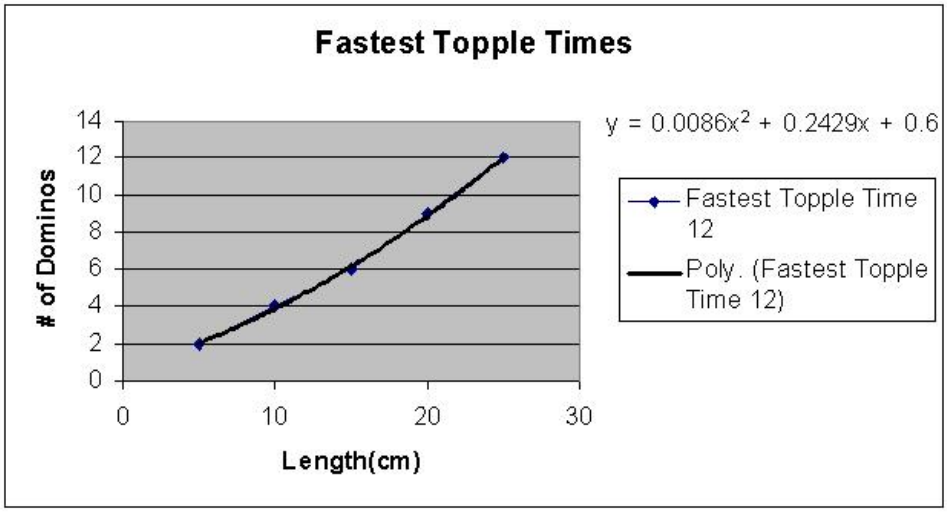
Chart 4

The minimum topple time of .555 seconds occurred with eight dominos and a spacing of 3.47 centimeters. However, with twelve dominos the topple time was .559 seconds, only a few thousandths of a second less then the value taken with just eight dominos. This number of dominos makes more sense when compared with the data shown at the other lengths. It seems as though the total topple times only slightly differ between eight and twelve dominos. In the following graph all of the fastest topple times are shown for each length along with the number of dominos used. Because of the varying times for twenty five centimeters all of the combinations between eight and twelve dominos are shown below.



Graph 4

As expected the trial done with twelve dominos seems to correlate with the previous trends from other trials. Using the graph of *Length vs. Number of Dominos* a third order polynomial fit was constructed.



Graph 5

Given a specific length it was now possible to predict the number of dominos necessary to minimize topple time. With "X" being equal to the given length in centimeters and "Y" being the number of dominos to choose, the following equation will be used:

$$Y = 0.0086 * X^2 + 0.2429 * X + 0.6 \tag{6}$$

For values of Y which are not whole numbers we simply round up or round down. However, this equation has a problem when the length that is plugged in is small. So we have to check to make sure the spacing makes sense before we use it. Now that a model had been created to minimize topple time with equal spacings, it was crucial to look at non-uniform spacings and figure out if our original prediction was correct.

Noting that there was not only one spacing which minimizes every string of dominos an idea was created to use the spacings for different sections of the sequence. Several different concepts were generated which enabled us to test our theory that equal spacings produced the fastest topple times. First, the ideal spacings for each of the appropriate sections was documented and the first and second spacings were averaged. The same thing was done with the final spacing and the previous one before it. Both of these averages were then also averaged and this average value was then compared with the data taken for equally spaced dominos at a given length. The number of dominos "N" was chosen by noting which trial had a spacing most similar to the average value calculated. Taking the first spacing as a reference value this number of dominos "N" was then plugged into the following equation and the end spacing ($L2$) was solved for:

$$D = (N - 1) * \frac{L1 + L2}{2} + 0.7 \tag{7}$$

- D = The Given Length
- N = The Number of Dominos
- $L1$ = Initial Spacing
- $L2$ = End Spacing

Keeping the first spacing ($L1$) constant, a linear progression was developed to decrease the spacing according to its slope until it equaled the end spacing ($L2$). The slope of this function was the following:

$$M = \frac{L1 - L2}{N - 2} \tag{8}$$

Similar tests were conducted simply changing the reference point from the first spacing to the last. Other experiments were done in the same manner but instead only an average of the exact first and last spacings was calculated. The results for these trials were the following:

Length(cm)	# of Dominos	start time	stop time	fall time	
15	5	1.030	0.582	0.448	* First spacing as reference
15	5	2.019	1.518	0.501	* Second spacing as reference
15	5	1.185	0.682	0.503	* Exact spacing values for first spacing
15	5	1.185	0.682	0.503	* Exact spacing values for last spacing

Chart 5

The topple time when using the first spacing as reference was less then that of the equally spaced case. This seemed to prove our original hypothesis wrong. However, just to make sure, the experiment for the fifteen centimeters of equally spaced dominos was conducted again. This time the equally spaced topple time was once again lower then the non-uniform case. Reasons for this were simply attributed to the fact that the tests were conducted on separate days and environmental conditions may have played a role in the slight change. None of the other non-uniform spacings provided faster topple times, reemphasizing that our equally spaced model was the best choice.

As with experiments done using the photogates, several sources of error were noted. Some errors were the same such as the error associated with lining the dominos up at their correct positions. However, these errors were minimized by having the same person line the dominos up each time. Similarly, there was large error associated with the reading of the initial domino release. As with the spacing, the operator remained constant and consistent results were achieved. As soon as the domino left the screw the sensor would read zero and the topple time would begin. Using the high speed camera allowed for other sources of error to be noted that could not be seen using the photogates. One such error was that often times the bottom of the domino would shift before falling and thus hit higher on the next domino. However, this is not necessarily a source of error that should be dismissed. Instead, it was used to improve the theoretical models to provide more accurate predictions.

Following the optimization of our model over any given length (using equidistant spacings), it was important to develop a theory for predicting topple times of any random array of dominoes. The only experimental restriction was that the set of dominoes remain in a straight line. The goal was to use large amounts of experimental data to predict the fall time for any string of dominoes with various spacings. Unlike our previous models, we found it very difficult to construct an experiment which would yield the necessary data to make an accurate prediction. This was simply due to the random nature of the arrangement. After struggling for a time, an experimental procedure was developed which would use the average spacing between the first 7 dominoes along with subsequent contact times (between dominoes) and fall time of the last domino to sum up the total topple time.

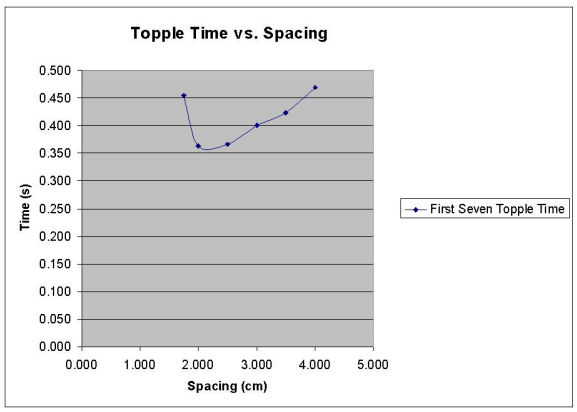
As with previous experiments, we used Shaw's assumption that energy transfer remains constant after the first six dominoes. However, Shaw's hypothesis only refers to equidistant spacings within a given range (2-3.5cm). In our case one additional assumption was necessary. After a few initial trials, it was concluded that the topple time for a random set up of dominoes was close to the topple time of another arrangement. This arrangement consisted of setting the dominoes up equally spaced at the average value of the random spacings. The topple time for these first six dominos was taken as the time it took for the first domino to leave the screw until the sixth domino came in contact with the seventh. As mentioned above this resulted in us averaging the spacings between the first 7 dominoes rather then the first 6 as one may expect. Trials were conducted at spacings of 1.75 cm, 2.00 cm, 2.50 cm, 3.00 cm, 3.50 cm, and 4.00 cm. For spacings

outside of this range our data becomes unclear. Averages for each spacing were calculated and the results were as follows:

Spacing (cm)	Average Topple Time (s)
1.750	0.454
2.000	0.363
2.500	0.367
3.000	0.401
3.500	0.423
4.000	0.468

Chart 6

It can clearly be seen that there is a minimum topple time somewhere between 2.00 and 2.5 centimeters. Graphically this minimum becomes even more apparent.



Graph 6

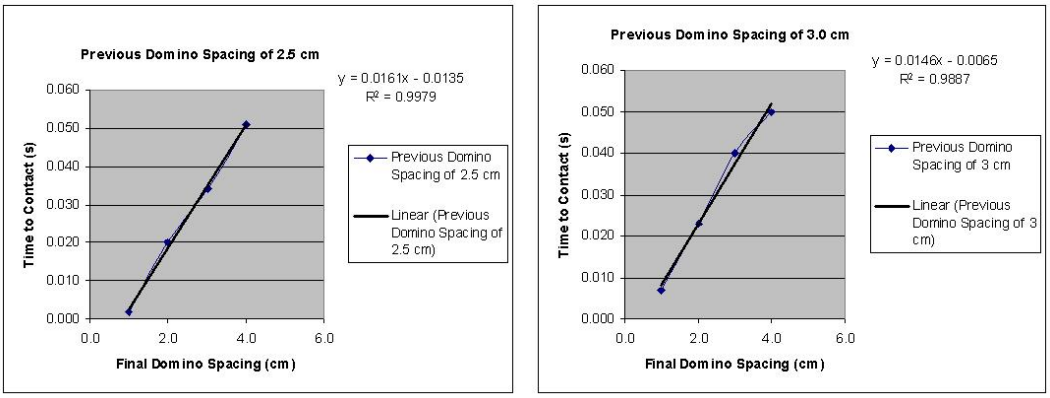
Now that a model for the first 6 dominoes had been developed it was important to establish a model for each of the subsequent dominoes. The idea behind this was to sum each of the contact times which could be interpolated from our experimental charts. These values would depend on the average spacing of all previous dominoes as well as lone spacing in front of the domino being measured. Because of the arbitrary spacing experiments were conducted at 1.0 cm, 2.0 cm, 3.0 cm, and 4.0 cm. This covered almost the entire range of possible combinations and resulted in the following:

		Average Previous Dominoes Spacing (cm)					
		1.75	2.0	2.5	3.0	3.5	4.0
7th-8th Domino Spacing (cm)	1.0	0.008	0.005	0.002	0.007	0.004	0.007
	2.0	0.031	0.026	0.020	0.023	0.023	0.016
	3.0	0.051	0.044	0.034	0.040	0.034	0.033
	4.0	0.050	0.048	0.051	0.050	0.045	0.052

Shaded=Contact Times (s)

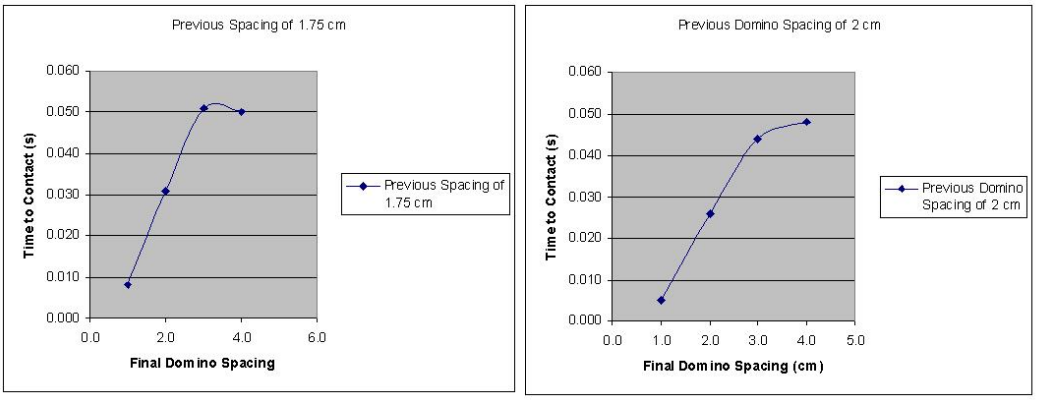
Chart 7

As expected the contact times increased when the spacing of the two relevant dominoes grew larger. In most cases, graphing the contact time versus the final domino spacing produced very linear lines. This is most clearly noted in the graphs for a previous spacing of 2.5 and 3.0 centimeters.



Graph 7

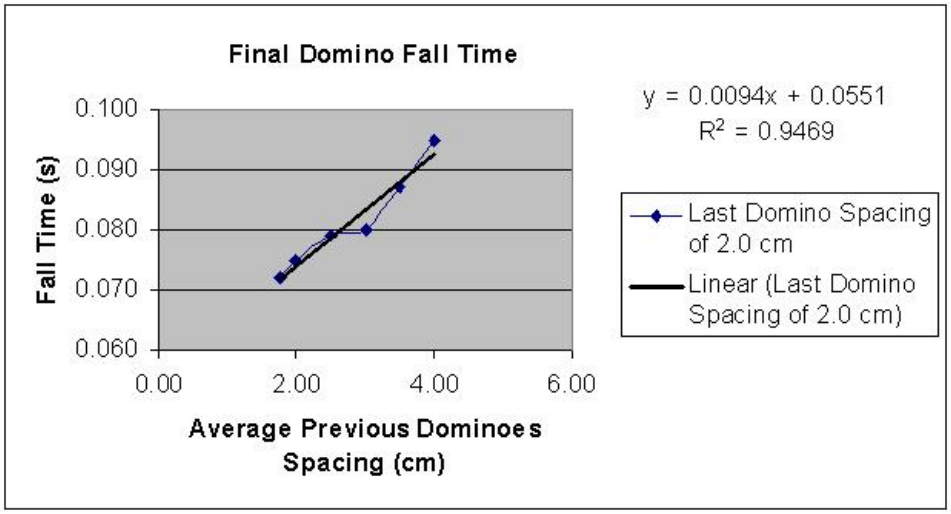
However, it was interesting to note that when the average previous domino spacing was less then 2.5 centimeters the results became vague. The graphs remained linear until a final domino spacing of 4.0 centimeters was measured, in which case the graph seemed to taper off as shown below.



Graph 8

This may simply be attributed to the scale at which these times are being measured. If each of these lines had remained linear through the 4.0 centimeter spacing the contact times would have only been altered by a total of less then a tenth of a second. This allowed us to neglect these errors and continue on with our experiments.

Similar results were shown when calculating the final domino fall time. As the average spacing of all previous dominoes became larger, the fall times also increased. In some cases a somewhat linear progression was noted as with the intermittent contact time graphs.



Graph 9

Once all of the data had been collected it was crucial that a method of interpolation be developed in order that all calculations would remain constant. The time it took for the first six dominoes to come in contact with the seventh was calculated through a simple interpolation. A sample trial was run in which the spacings between each of the first seven dominoes was measured using calipers and averaged. A spreadsheet was generated using Microsoft Excel to calculate these averages along with the values of interpolation. Once this average value was obtained, a standard interpolation was conducted in order to solve for an "x" value. This "x" value represented the position of this average relative to the data values which surrounded it. Once this "x" value was calculated it was possible to carry out another interpolation. This interpolation simply solved for the actual topple time of the first six dominoes by plugging in the correct "x" value from our spreadsheet as well as the upper and lower limits of interpolation. For a sample run of 10 randomly spaced dominoes the results were as follows:

First Seven Dominoes	
Average Spacing (cm)=	2.4366667
$x_{1.75\text{-}2}$ =	2.7466667
$x_{2\text{-}2.5}$ =	0.8733333
$x_{2.5\text{-}3}$ =	-0.1266667
$x_{3\text{-}3.5}$ =	-1.1266667
$x_{3.5\text{-}4}$ =	-2.1266667
Input X Value	0.8733333
Lower Limit	0.363
Upper Limit	0.367
Time To Contact Domino 7=	0.366493333

Shaded=Inputs

Chart 8

Now that we had a time for the first six dominoes it was important to calculate the contact times for all of the following dominoes as well as the fall time of the final domino. To do this a similar pattern was followed.

All of the subsequent averages for previous domino spacings were calculated through the spreadsheet and then the relevant "x" values were solved for. However, in the case of all dominoes after 6 it would require multiple interpolations. This was because an interpolation would also be necessary for the spacing relevant to the two dominoes being evaluated. The same would be true for the topple time of the final domino. After all the contact and fall times were calculated, they were added to the initial time of the first six dominoes. Compared with the time taken from the Photron Motion Tools program our test trial yielded an error slightly less then 5% as shown below.

Experimental Time (s)	0.49980
Photron Motion Tools Time (s)	0.52600
% Error=	4.98099

Chart 9

Several other trials were conducted in which most errors were similar to the one previously mentioned. However, in the case of only eight dominoes an error of 30% was noted. Obviously this error is too great and must be fixed. More trials were run to further explain this huge jump in error.

As with all other experiments there are several sources of error which must be noted. The most notable error once again was the human error in measuring when exactly the domino leaves the screw in Photron Motion Tools. An additional inaccuracy which was noticed, was relative to the plastic screw compressing. A closer look at the videos seems to reveal that the screw compresses a bit when pushing the domino forward. Initially, this error would be seen as irrelevant but because of the error involved in reading when the domino leaves the screw it may be significant. It is possibly that the screw may be actually pushing the domino through the first few small angles of rotation. This may be a very important discovery for our theoretical model and thus will be noted.

Following the transformation of our theory to accommodate for non-uniform spacings we began to look at the effect of non-linear arrangements. As expected this was a very difficult task. In order to account for curved line dominoes one must also account for the rotation which takes place as dominoes hit on an angle. In theory this should cause the system to lose energy and the overall topple times to increase. Because of the random nature which a non-linear string of dominoes exhibits, it was important to come up with a testing plan.

Initial tests were conducted in which only two dominoes were set up. The first domino was set straight up and the second was simply tilted on an angle. The spacing was taken as the distance from the front face of the first domino to the corner at which the domino pivots. In the case of the angles, an imaginary line was drawn parallel to the back face of the previous domino. This line extended from the corner of the second domino outwards. The angle was then measured from this line to the front face of the second domino. A schematic of these measurements is shown below.

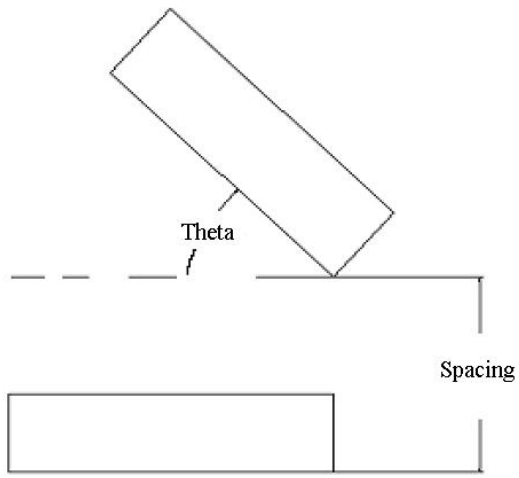


Figure 1: Measurement Setup

Data was collected at angles of 0°, 30° and 60° degrees. Looking at the data seems to indicate that as the angle is increased the topple time increases as well. The reason for this could be that not as much energy is transferred as when the dominoes strike face to face. Also, if the angle is great enough only the corner of the falling domino will be transferring energy. The resting angles of each domino were also tabulated but no significant information was gained from them.

x (cm)	y (cm)	θ (deg)	time (s)	Resting Angle 1(deg)	Resting Angle 2(deg)
2	0	0	0.307	0	0
2	0	30	0.311	30	30
2	0	60	0.357	40	20
3	0	0	0.361	0	-25
3	0	30	0.376	10	30
3	0	60	Hit, but No Topple	30	0

Chart 10

An experiment was also developed in which the second domino would be displaced in the y-direction as well as have angle changes. A schematic of this setup is shown below.

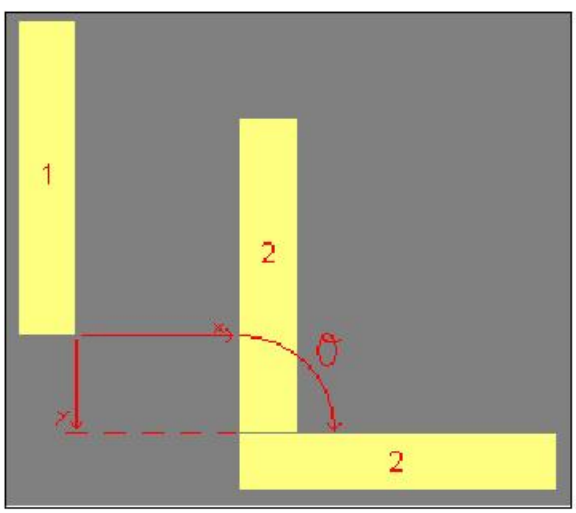


Figure 2: Schematic of Setup

Tests were run at the same angles in which the dominoes were displaced one quarter of the length and one half of the length of the domino. This time the experimental data did not seem to lean towards the idea of topple times increasing as the angles increased. It was also interesting to note that some of the dominoes fell backwards and in some cases the dominoes never contacted.

x (cm)	y (cm)	θ (deg)	time (s)	Resting Angle 1(deg)	Resting Angle 2(deg)
2	0.539	0	0.312	5	10
2	0.539	30	0.373	5	35
2	0.539	60	0.357	-20	60
3	0.539	0	0.348	5	10
3	0.539	30	0.335	-5	20
3	0.539	60	Fell Backwards	n/a	n/a
2	1.078	0	0.307	-10	40
2	1.078	30	0.315	-5	60
2	1.078	60	No Contact	n/a	n/a
3	1.078	0	0.292	0	0
3	1.078	30	0.335	-10	60
3	1.078	60	No Contact	n/a	n/a

Chart 11

Unlike the two domino case, experiments were also conducted with larger strings of dominoes. First the dominos were set up along curves which were semicircular in shape. All spacings and angles were measured exactly the same as with the two domino case. A metric ruler was used to measure the spacings and a protractor was used to approximate the various angles along the curve. Because of the location of the camera a new arrangement was necessary for conducting these tests. The following is a picture of a typical semicircular setup.

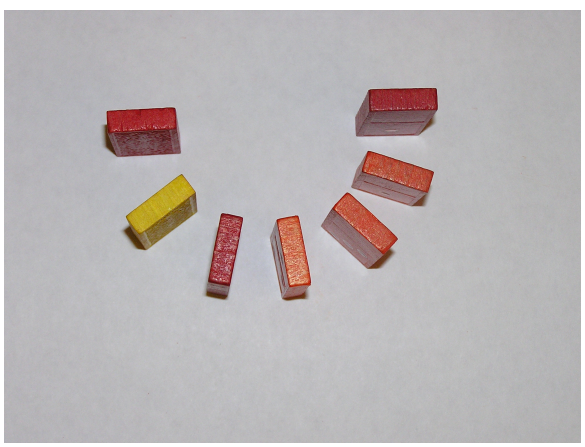


Photo 2: Dominoes in Semi-circle pattern

As expected, the topple times increase relative to comparable strings of straight dominoes. In the case of 6linear dominoes equally spaced at 2.0 centimeters the average topple time was 0.363 seconds.

	Spacing(cm)	*Start Time(s)*	*Stop Time(s)*	*Topple Time(s)*
Trial #1	2.000	1.817	2.172	0.355
Trial #2	2.000	2.061	2.428	0.367
Trial #3	2.000	1.774	2.123	0.349
Trial #4	2.000	1.904	2.284	0.380

Average Topple Time (s)=	0.363

Chart 12

However, when a curved string of 6 dominoes with an average spacing close to 2.0 centimeters was tested, a much different result was noted. For this experiment the topple time nearly doubled at 0.623 seconds.

This increase in time is due to the energy lost in the system. Often times the entire face of the dominoes do not contact and thus a rotation occurs. These rotations are where much of the energy goes. Also, it is important to note that the energy transferred throughout the system does not become constant until after the $6th$ domino. Some variations in the data may be attributed to this concept.

Similar to the semicircular curves, an arrangement was setup which resembled a bell curve. All measurements were still gathered in the same manner as with previous tests. However, in this case negative angles will be measured. This occurs due to the interesting geometry of the setup.

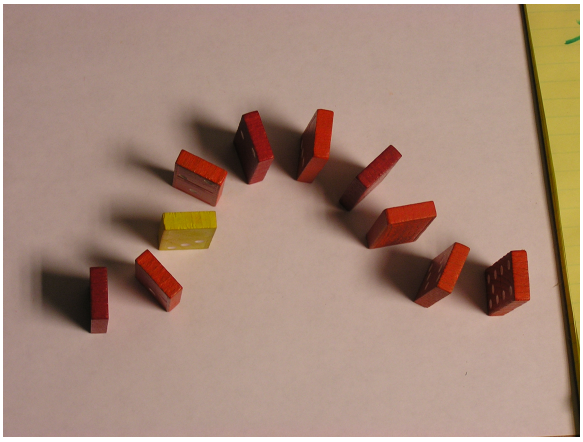


Photo 3: Dominoes in Bell pattern

As with the semicircular setup the topple times increased by a relatively large amount, yet there were no patterns noticed during the different trials. This is likely due to the many different factors now affecting our system such as orientation and energy loss. In a previous linear experiment of eight randomly spaced dominoes the topple time was measure to be 0.678 seconds. The average spacing of the dominoes in this trial was 2.206 centimeters. Using the bell shaped curve with an average spacing of 2.26 centimeters the time increased nearly 30% to 0.888 seconds.

In both the case of the semicircular setup as well as the bell shaped curve, it may be possible to optimize a topple time over a given length. The reason for this is simple. If the length and number of dominos are set variables then it may not be possible to fit all of the dominoes in a straight line. Even if the dominoes can fit closely together they may topple at a much slower rate because of their proximity to one another. In cases such as this, one of the previously mentioned curved setups may be the best option.

It becomes obvious that the angle at which the dominoes are placed influences the topple time greatly. Further corrections to our theory have been made in order to solve this difficult issue. Comparisons of the previously mentioned experimental values and theoretical predictions will be addressed in the following sections of the report.

4 Theoretical

Our theoretical model has gone through a good amount of revision. For the newest revision, we went back to the equation for the conservation of energy. This equation is:

$$E = (\frac{1}{2})mg(b\cos\theta + a\sin\theta) + \frac{1}{2}I\dot{\theta}^2, \tag{9}$$

where $I = \frac{1}{3}m(a^2 + b^2)$ is the moment of inertia, a is the thickness of the domino, b is the height of the domino, m is the mass of the domino, g is the acceleration due to gravity, θ is the angle of displacement from the vertical, and $\dot{\theta}$ is the angular velocity.

This equation was differentiated to obtain:

$$0 = (\frac{1}{2})mg(a\cos\theta - b\sin\theta) + I\ddot{\theta}. \tag{10}$$

We wished to simplify this equation by applying dimensional analysis. So we let $T = \frac{t}{\tau}$. Which gave us the relation $\frac{d^2\theta}{dt^2} = \frac{1}{\tau^2}\frac{d^2\theta}{dT^2}$. This was substituted into the above equation. After the substitution, we decided to let $\tau = \sqrt{\frac{2I}{mgb}}$. This gave us the equation:

$$\ddot{\theta} - \sin\theta + \frac{a}{b}\cos\theta = 0. \tag{11}$$

We now had our differential equation with initial conditions $\dot{\theta}(0) = \dot{\theta}_0$ and $\theta(0) = \theta_0$. How $\dot{\theta}_0$ and θ_0 are chosen will be discussed later. The above equation was multiplied through by $\dot{\theta}$ to get:

$$\dot{\theta}\ddot{\theta} + \dot{\theta}(\frac{a}{b}\cos\theta - \sin\theta) = 0. \tag{12}$$

Which is equivalent to

$$\frac{d}{dT}(\frac{\dot{\theta}^2}{2} + (\frac{a}{b}\sin\theta + \cos\theta)) = 0. \tag{13}$$

Then when we integrate we obtain:

$$\frac{\dot{\theta}^2}{2} + (\frac{a}{b}\sin\theta + \cos\theta) = E. \tag{14}$$

Where E is the energy of the system found with our initial conditions. This equation is now separable. So we can obtain:

$$\int_{\theta_0}^{\theta_f} \frac{d\theta}{\sqrt{2}\sqrt{E - (\frac{a}{b}\sin\theta + \cos\theta)}} = \int_{T_0}^{T_f} dT. \tag{15}$$

This should give us the time for one domino to fall from θ_0 to θ_f. This will be in the non-dimensional variable T so it will need to be multiplied by τ to get the time in actual seconds.

From this model we wrote a program that could calculate the topple time of N dominoes at a specific spacing. We did this by creating a system that would make new initial conditions for each domino. Using our non-dimensional equation, we found the angular velocity of domino$_i$ as it hit domino$_{i+1}$ by simply plugging in the angle θ_f which can be found from geometry. The velocity of the domino is tangential to the angle of the domino. So we took the horizontal component of the velocity and used it in the equation for conservation of momentum $m_n v_{n,f} + m_{n+1}v_{n+1,0} = (m_n + m_{n+1})v_{n+1,f}$ where $v_{n+1,f}$ is the new initial velocity after collision for domino$_n + 1$. Using our assumption $v_{n+1,0} = 0$, this becomes $\frac{m_n}{m_n+m_{n+1}}v_{n,f} = v_{n+1,f}$.

This model does not agree as well with our experimental data as we had hoped. So we decided to write a new model. The motion of a single domino is very similar to the motion of a simple pendulum. The motion of a pendulum can be found by the differential equation

$$\ddot{\theta} + \frac{g}{l}\sin\theta = 0, \tag{16}$$

where $\ddot{\theta}$ is the angular acceleration, g is the gravitational constant, l is the length from the axis of rotation to the center of gravity of the domino, and θ is the displacement from the negative vertical axis where the origin is the axis or rotation.

Since the domino is basically an inverted pendulum, we changed the equation to $\ddot{\theta} + \frac{g}{l}\sin(\theta + \pi) = 0$. But since $\sin(\theta + \pi) = -\sin\theta$, the equation became $\ddot{\theta} - \frac{g}{l}\sin(\theta) = 0$, where θ is now the displacement from

the positive vertical axis. We also took l to be the length from the axis of rotation to the center of mass. We assumed this distance to be $\frac{1}{2}d = l$, where d is the diagonal length of the domino.

Using the same substitution in out previous model, $T = \frac{t}{\tau}$ and $\frac{d^2\theta}{dt^2} = \frac{1}{\tau^2}\frac{d^2\theta}{dT^2}$, we have $\frac{1}{\tau^2}\ddot{\theta} = \frac{g}{l}\sin\theta$. So we let $\tau = \sqrt{\frac{g}{l}}$ and we now had $\ddot{\theta} = \sin\theta$.

Using similar manipulations in our previous model, we came up with

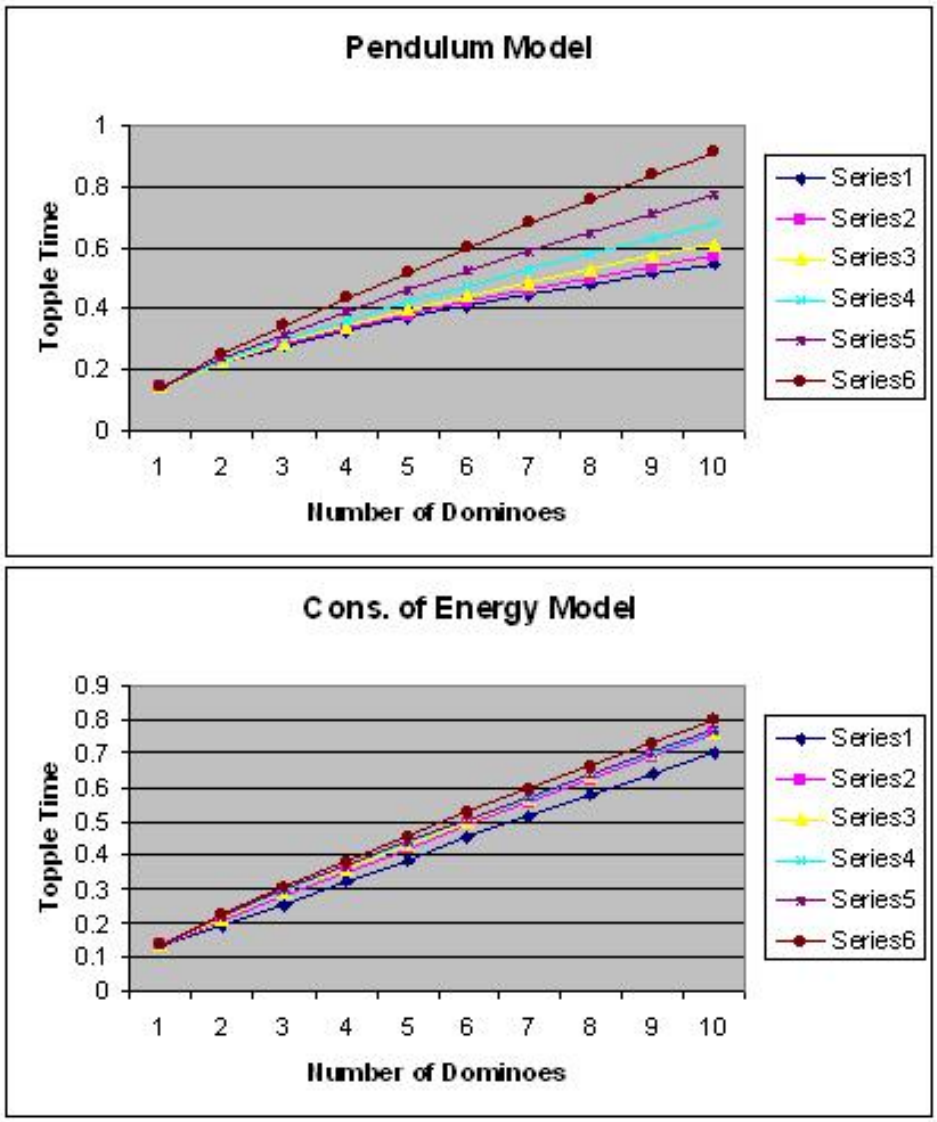
$$E = \frac{\dot{\theta}^2}{2} + \cos\theta. \tag{17}$$

Now to get the topple time of one domino, we can separate variables and integrate to get,

$$\int_{\theta_0}^{\theta_f} \frac{d\theta}{\sqrt{2}\sqrt{E - \cos\theta}} = \int_{T_0}^{T_f} dT. \tag{18}$$

Multiplying T_f by the new τ will give us the new topple time for one domino.

We wrote a program similar to the one for our previous model so we could find the total topple time for n dominoes at spacing h. The new theoretical data that this gave us agreed better with our experimental data. This figure shows the topple times that were found using this model and our program.



Graph 10: Theoretical Graphs for Topple Times

5 Comparison

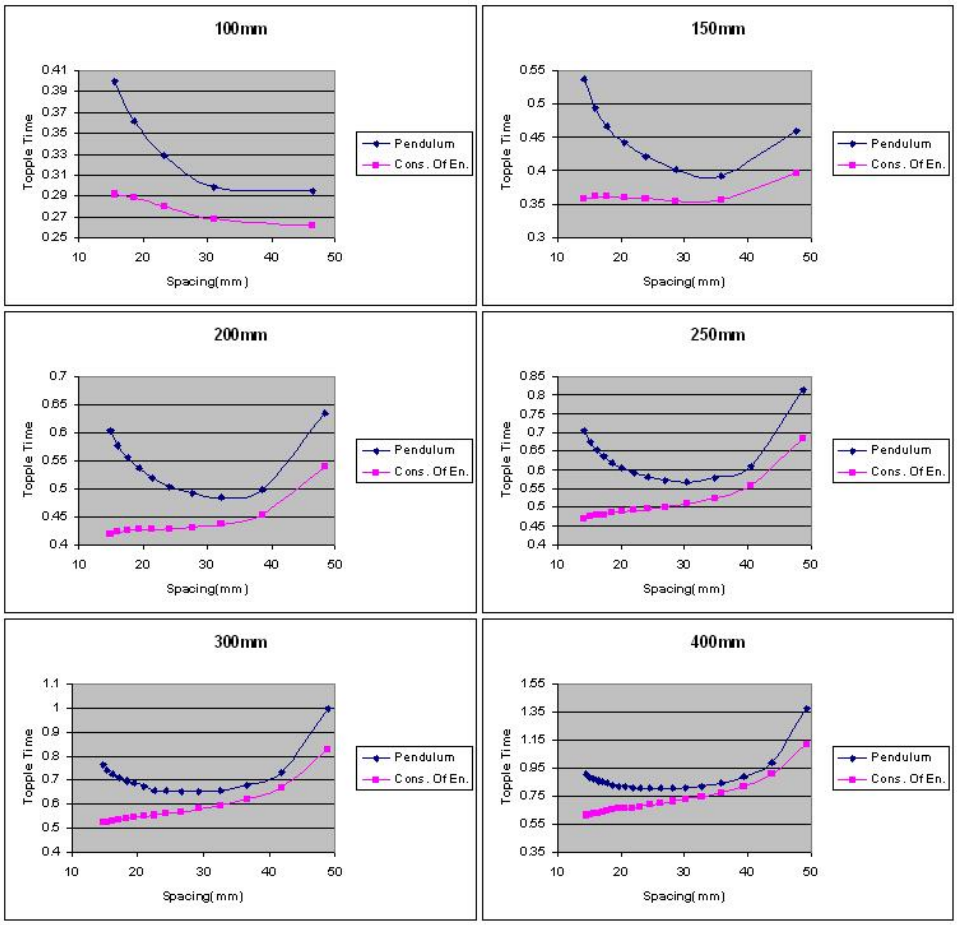
Using our conservation of energy model, we had an average relative error of 15.4%. Our pendulum model had an average relative error of 7.5%. Comparing these to an average relative error of 0.9% for our linear fit, we can see that our theoretical model is getting better. Even though our linear fit appears to be the best estimation for topple time, we learned from competition one that it was actually a poor estimation.

This is most likely because the relative error for the linear fit is somewhat deceiving. The equation for the linear fit was found from the data obtained experimentally for between six and ten dominoes each at spacings of 1.5cm, 2.0cm and 2.5cm. So the relative error obtained using the linear fit will be small when it is within the interval of six and ten dominoes. Since extrapolation is very dangerous, applying this linear

fit to values outside these intervals will most likely result in much larger relative errors. The graph above shows the linear fit line over the interval of one to ten dominoes. The difference between the linear fit and our models at spacings greater than 2.5cm is quite large.

For competition two, we had to minimize the topple time for a given length L. In our experimental section we showed that we found an equation to estimate the spacing needed for a given length. So we took this and compared it what we found with our pendulum model.

We took both models and applied it to different lengths with different spacings for each. In this model, all of the dominoes are equally spaced. This figure shows the topple times found for the different lengths as it varied over domino spacing.



Graph 11

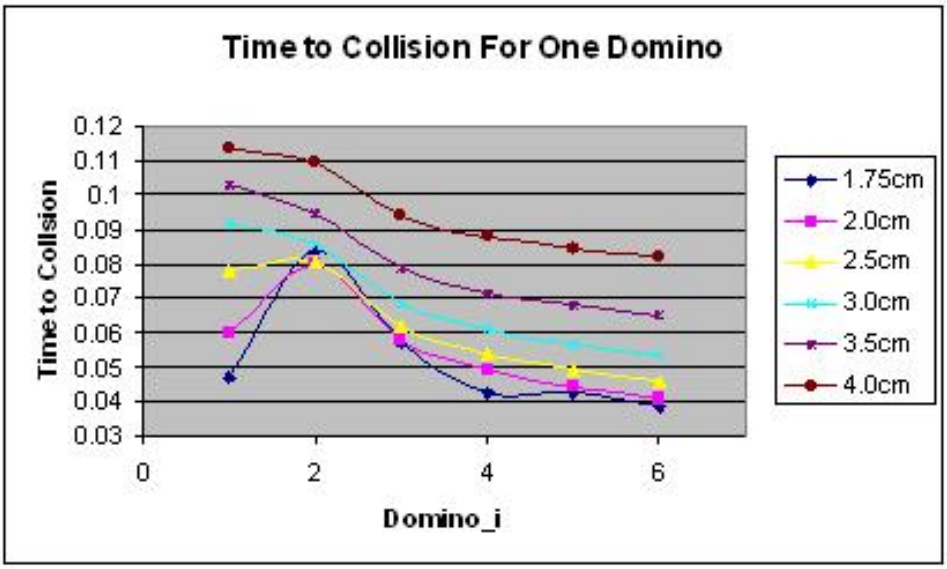
Against our experimental data, these graphs compare well for shorter lengths. Experimentally, we found the best spacing to be as large as possible. Theoretically, this matches up with what we found for 10 cm and 15cm. However, as the length increases our theoretical model shows that decreasing the space between the dominoes yields faster topple times. Compared against the topple times found over a length of 25cm, our Pendulum Model had an average relative error of 6.0 % and our Conservation of Energy Model had an average relative error of 15.8%. This gives us a little more evidence that our pendulum is a better predictor.

It is also interesting to note how differently the models behave. Our Pendulum Model appears to suggest that at each length greater than 10cm there is a specific spacing that would give the fastest topple time

for that length. Also, these spacings appear different at each length. If these spacings are different at each length, that would suggest non-equal spacing would give the fastest possible topple time.

The Experimental section explains how we made a new experiment where we had six trials of seven dominoes that had different equal spacings for each trial. Then for each trial, there were four tests run where the spacing between the seventh and eighth had spacings of 1.0cm, 2.0cm, 3.0cm, and 4.0cm. We decided to apply this same experiment against our Pendulum Model.

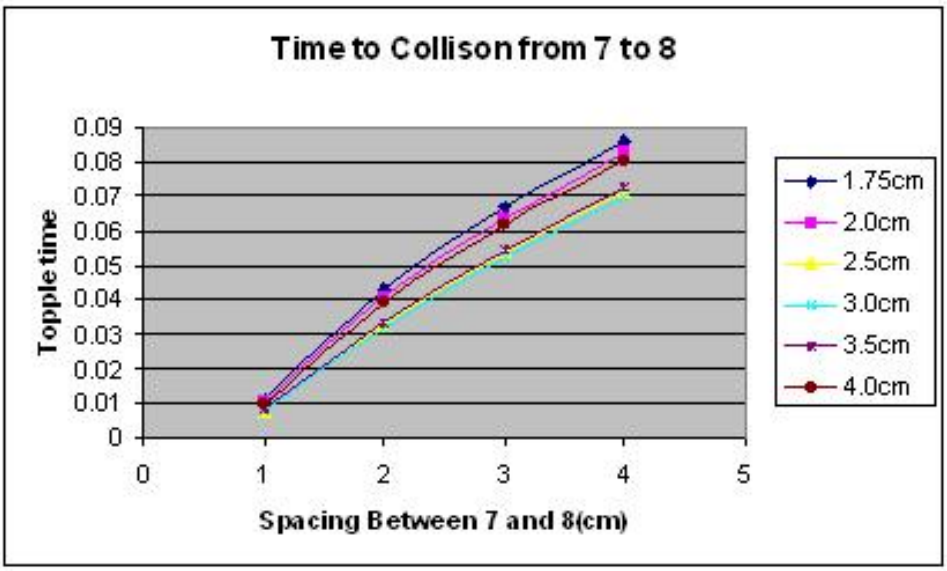
This graph shows the time to collision at each domino for each spacing.



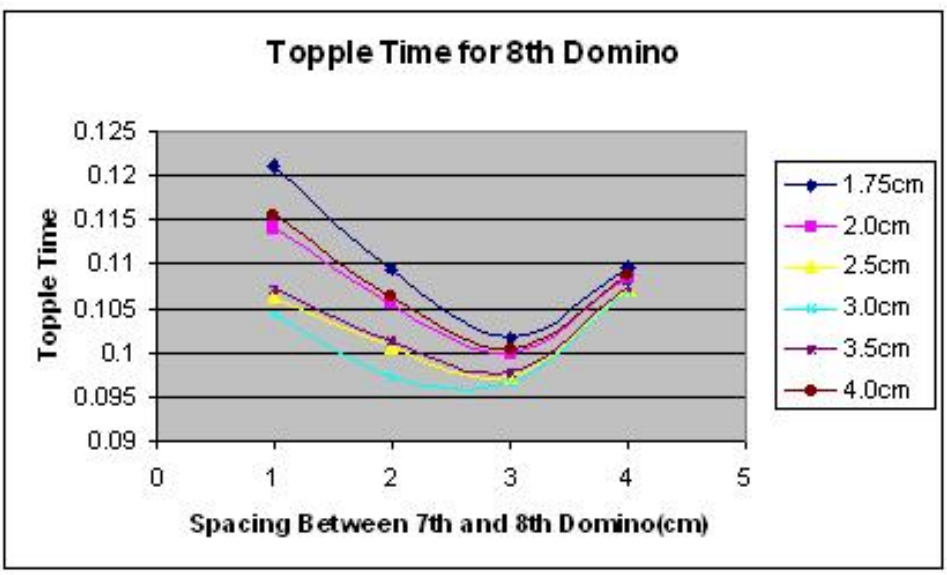
Graph 12

What is interesting about this graph is at Domino_2 the fastest time to collision between it and the next domino is when the spacing between it and the previous domino is 2.0cm. Then after that domino the spacing that gives the fastest time to collision is 1.75cm. This gives more evidence to the theory that non-equal spacing would give the fastest topple time. At the same time, this raises more questions about how we model the transfer of momentum between dominoes.

These next graph show the time to collision between the seventh and eighth domino and the topple time for the eighth domino.



Graph 13



Graph 14

The first graph shows the time to collision is the smallest when the spacing between the seventh and eighth is 1.0cm, which makes sense because this is also the smallest of the spacings. What is interesting is at each different spacing between the seventh and eighth, the fastest time to collision is when the spacing between the first seven is 3.0cm. The second graph shows the time to fall for the eighth domino is fastest when the spacing between the first seven is 3.0cm and the spacing between the seventh and eighth is also 3.0cm. This makes the inference that after a certain number of dominoes there is specific spacing that would give the fastest topple time. However, the same questions about the transfer of momentum are raised.

We then compared these results with our experimental data. The average relative error between the times found from when the first domino falls to when the sixth hits the seventh was 13.2%. The difference between the times found from when the sixth hit the seventh and the seventh hit the eighth was an average

relative error of 68.5%. This was somewhat scary. But realizing that the actual differences were on the scale of thousandths of a second, we did not take this scare seriously. The average relative error between the times found for the topple time of the eighth domino was 34.2%. The actual differences for these were on the scale of hundredths of a second, so this was taken a little more seriously. The average relative error for the total topple time of the eight dominoes was 13.3%.

The next problem we wished to address was when the line of dominoes is not a straight line. Our current model did not account for this, so we decided to modify it. We came to the conclusion that we would need to incorporate a term into our model for the transfer of momentum that included the angle of displacement between a line parallel to the front of domino$_i$ and a line parallel to the back of domino$_{i+1}$. We took our previous model for $v_{i+1,0}$ and now made it $\frac{m_i}{m_i+m_{i+1}} v_{i,f} w \cos\phi = v_{i+1,0}$, where ϕ is the angle of displacement that was just mentioned.

Another problem with this setup was what to define as the distance between two dominoes. We used three different ways to define the distance and tried to see if one was consistently better at estimating the topple time. These ways were measuring from the front of domino$_i$ to the front corner of domino$_{i+1}$, the front of domino$_i$ to the middle of domino$_{i+1}$, and the front of domino$_i$ to the back corner of domino$_{i+1}$. So now instead of just the horizontal component of $v_{i,f}$, the model involves the component that is normal to the face of domino$_{i+1}$. When we compared our theoretical times to our experimental times for the semi-circle setup, it turned out that for each trial the best estimation came when the distance between dominoes was defined as the distance to the back corner of domino$_{i+1}$. The average relative error was 14.1% when the distance was measured to the back corner, 23.2% when it was measured to the middle, and 27.6% when it was measured to the front corner. All of these errors were higher than we would like. But when taking into consideration the difficulty and experimental error when measuring the angles of displacement between dominoes, we decided this was the best obtainable result.

6 Discussion

One interesting result came up when we examined the graphs for the pendulum and conservation of energy models. For fewer than six dominoes, the topple times do not appear to grow linearly as the number of dominoes increase. After six dominoes, the increase in topple time appears much more linear as the number of dominoes increase. This compares very well with Shaw's observation that the energy of the system becomes constant when the number of dominoes is greater than six.

This does however expose some flaws in both of our models. The energy of the system at domino$_n$ was taken to only be the sum of the potential and kinetic energy of domino$_n$ at its initial conditions. This does not take into consideration the energy from domino$_{n-1}$ or further back. This means at domino$_i$, the energy is constant but possibly different than the energy at domino$_j$ depending on the initial conditions at i and j, where $i \neq j$.

We attempted to correct this by changing how the energy at domino$_n$ was computed in our program. The article by Shaw showed how θ_n related to θ_{n-1} and how $\dot{\theta}_n$ related to $\dot{\theta}_{n-1}$. The equations were $\sin(\theta_{n-1} - \theta_n) = (c\cos\theta_n)/b - \frac{a}{b}$ and $\dot{\theta}_{n-1} = (1 - (c\sin\theta_n)/b\cos(\theta_{n-1} - \theta_n))\dot{\theta}_n$.

Using this recursive relation, we went back to our pendulum model and made

$$E = \frac{1}{2}\sum_{1}^{n} \dot{\theta}_i^{\,2} + \sum_{1}^{n} \cos\theta. \tag{19}$$

However, this did not work for n greater than 4.

Another flaw that was shown was how $\dot{\theta}_{n+1,0}$ is chosen in our program. We said that we chose $v_{n+1,0} = \frac{m_n}{m_n+m_{n+1}} v_{n,f}$ where $v_{n,f}$ is the horizontal component of $\dot{\theta}_{n,f}$. In this model we took m_n to be the sum of

the mass of dominoes from one to n and m_{n+1} to be the mass of one domino. Since we assumed that the mass of the dominoes were uniform, this becomes $v_{n+1,0} = \frac{n}{n+1} v_{n,f}$.

We also made the observation that where domino$_n$ hit domino$_{n+1}$ would effect the total amount of momentum that was transferred. We assumed this would be a ratio of the height of collision over the height of the domino. So the initial condition was now $\dot{\theta}_{n+1,0} = \frac{h}{b} v_{n+1,0}$ where h is the height of collision found by geometry and b is the height of the domino.

We decided we should investigate how the actual difference between $\dot{\theta}_{n,f}$ and $\dot{\theta}_{n+1,0}$ compared with how they differed in our model. To do this we went back to our videos of the toppling of seven dominoes at equal spacings of 1.5cm, 2.0cm, and 3.5cm. Around the collision of domino$_i$ and domino$_{i+1}$, we took three pictures. The first was five frames before the collision, the second at the collision, and the third five frames after the collision. Then for each picture we measured the angles from lines parallel to the fronts of the dominoes to a line parallel to the surface of the counter.

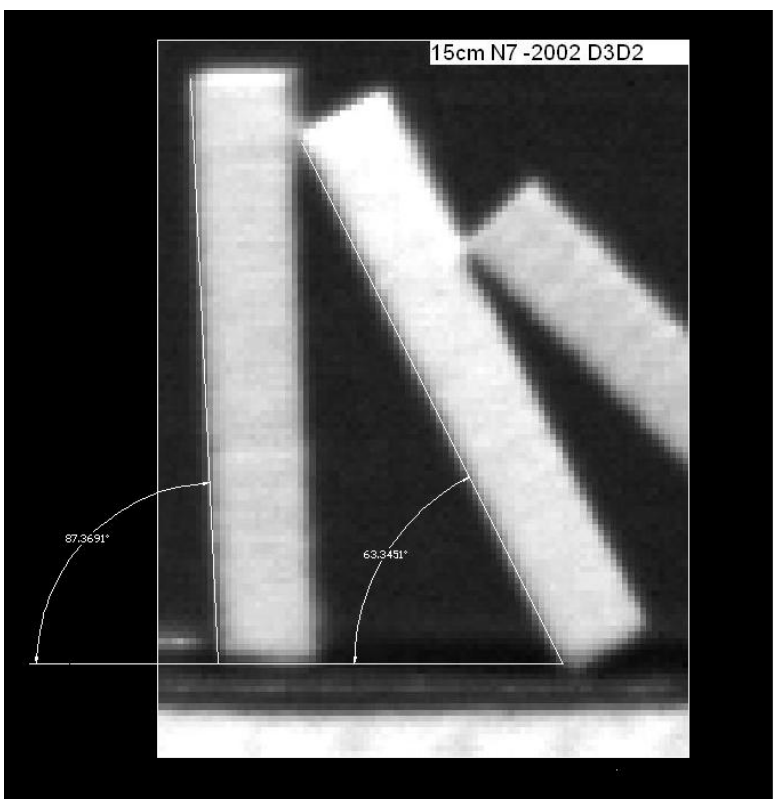
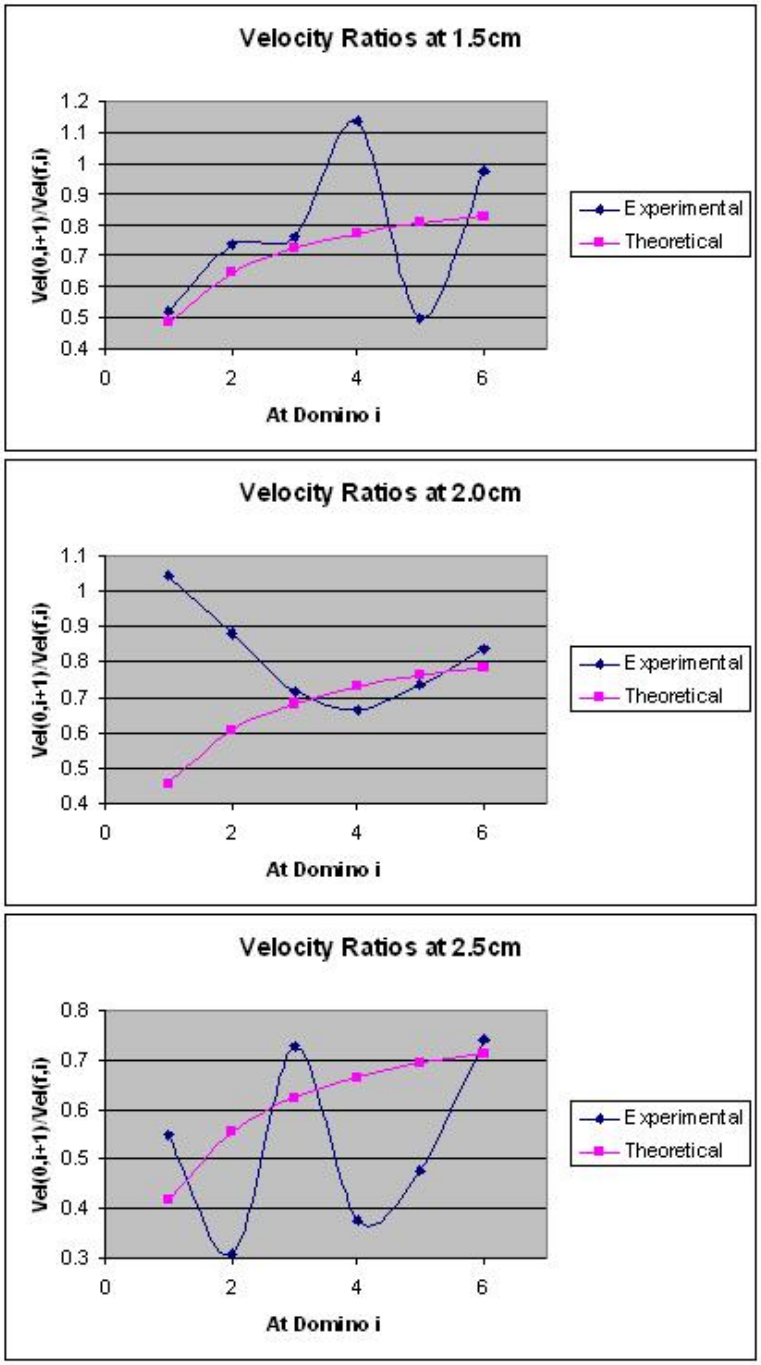


Photo 4: Frame from video used to record topple time, as well as angles used to measure velocities.

We then subtracted the angle of domino$_{i+1}$ at the third picture from the angle of domino$_{i+1}$ at the second picture and we subtracted the angle of domino$_i$ at the second picture from the angle of domino$_i$ at the first picture. Since the difference between the pictures was five frames and each frame had a difference of one thousandth of a second, we divided both of these by .005. This gave us our estimation for $\dot{\theta}_{n+1,0}$ and $\dot{\theta}_{n,f}$ respectively. We then got a velocity ratio for each collision that was equal to $\frac{\dot{\theta}_{n+1,0}}{\dot{\theta}_{n,f}}$.

Then we went back to our pendulum model and found the same velocities over the same range. We made the same velocity ratios and these were the graphs for each spacing of 1.5cm, 2.0cm, and 2.5cm.



Graph 15

The velocity ratios from our experimental data do not behave nearly as nice as our theoretical data but this is most likely to be expected. First of all, $\dot{\theta}_{n+1,0}$ and $\dot{\theta}_{n,f}$ were only estimations. Also there was most likely some human error in judging where the lines parallel to the front of the domino and the counter were drawn. All of this aside, the values were reasonably close. If we went back to other videos and found $\dot{\theta}_{n+1,0}$

and $\dot{\theta}_{n,f}$ at the same collisions and made an average of these velocities at the same collisions, we might find the error between our experimental and theoretical data to be better.

One part of the final competition was another minimizing topple time problem. However, this time we were given the total length of the line of dominoes and the number of dominoes that had to be in the line. So we went to our theoretical model and started with just two dominoes. We varied the distance between the two dominoes to find the distance that gave the fastest topple time. We found that a spacing of 20.1mm gave the fastest time.

The next part of this problem was what happens when there are only three dominoes. So we started with fixing the distance between the first two dominoes at 20.1mm and varied the distance between the second and third dominoes. In this setup, we found the fastest time came when the distance between the first two dominoes was 7mm or essentially when the second two dominoes were touching. This was a little discouraging because it could possibly show a flaw in our model.

We then went back and fixed the distance between the second two dominoes at 7mm and varied the distance between the first two dominoes. For this setup, we found that a spacing of 19.2mm between the first two dominoes gave the fastest topple time. We hoped that this distance would be the same we found in the first setup, but unfortunately it was not.

So next we wanted to see what would happen with four dominoes. We fixed the distance between the second and third dominoes at 7mm and the distance between the third and fourth dominoes at 7mm as well. When we varied the distance between the first and second dominoes, we found that a spacing of 18.5mm gave the fastest topple time.

All of this is a little discouraging because it does not make much sense that when more dominoes are added to the line, the fastest topple time comes when the distance between the first two dominoes was decreased. When the distance between the dominoes is decreased, the first domino has less time to accelerate which means $v_{1,f}$ will be less and therefore so will $v_{i+1,0}$. Also if every time a domino is added requires the distance between the first two dominoes to be shortened, there might be a point where the distance is not large enough for the first domino to topple.

References

[1] Banks, R. B. (1998).Towing IceBergs, Falling Dominoes, and Other Adventures in Applied Mathematics. Princeton, Princeton University Press.

[2] Bert, C. W.(1986)."Falling Dominoes."SIAM Review **28**(2):219-224.

[3] Heinrich, A.J., C.P. Lutz, et al. (2002). "Molecule cascades." Science **298**: 1381-1387

[4] Lin, Y., M. L. Hulting, et al. (2004). "Causes of spatial patterns of dead trees in forest fragments in Illinois." Plant Ecology **170**: 15-27

[5] MacAyeal, D.R., T.A. Scambos, et al.(2003). "Catastrophic ice-shelf break-up by an ice-shelf-fragment-capsize mechanism." Journal of Glaciology **49**(164).

[6] McGeer, T. and L.H. Palmer(1989). "Wobbling, Toppling, and Forces of Contact." American Journal of Physics **57**(12):1089-1098.

[7] Perkins,S. (2002). "Toppling icebergs sped breakup of Larsen B ice shelf." Science News Online **162**(25/26).

[8] Sachtjen, M.L., B.A. Carreras, et al. (2000). "Disturbances in power transmission systems." Physical Review E **61**(2): 4877-4882.

[9] Shaw, D.E. (1978). "Mechanics of a chain of Dominoes." American Journal of Physics **46**(6):640-642.

[10] Stronge, W.J. (1987)."The domino effect:a wave of devastating collisions in a periodic array." Proceedings of the Royal Society **409**:199-208.

[11] Stronge, W.J. and D. Shu (1988)."The domino effect: successive destabilization by cooperative neighbors." Proceedings of the Royal Society **418**:155-163.

[12] Wagon,S.,A. Pontarelli, et al.(2005)."The Dynamics of Falling Dominoes." Pre-print.

[13] Watts, D.J.(2002)."A simple model of global cascades on random networks." Proceedings of the National Academy of Science **99**(9):5766-5771.